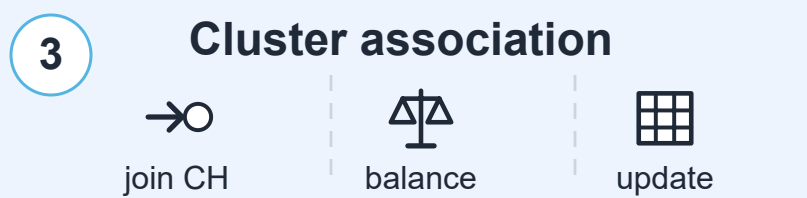
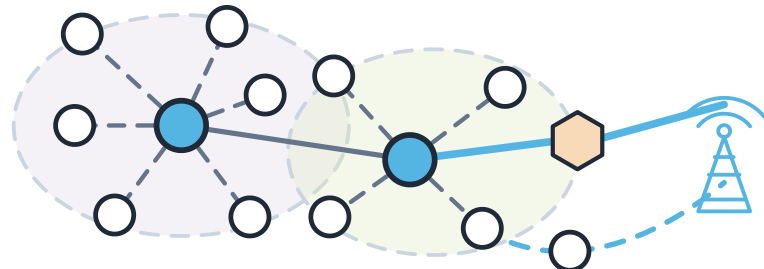


AERIS round pipeline



Phase 1 Cluster Formation



Phase 2 Intra-Cluster Communication

Inputs: energy · link quality · density · buffer

1 CAS mode selector

select one mode

Direct

M → CH



Chain

M → M → CH



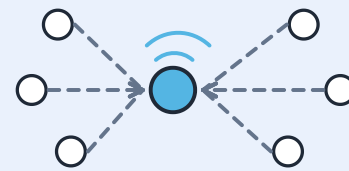
Two-hop

M → R → CH

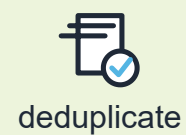


M R CH

2 Intra-cluster forwarding

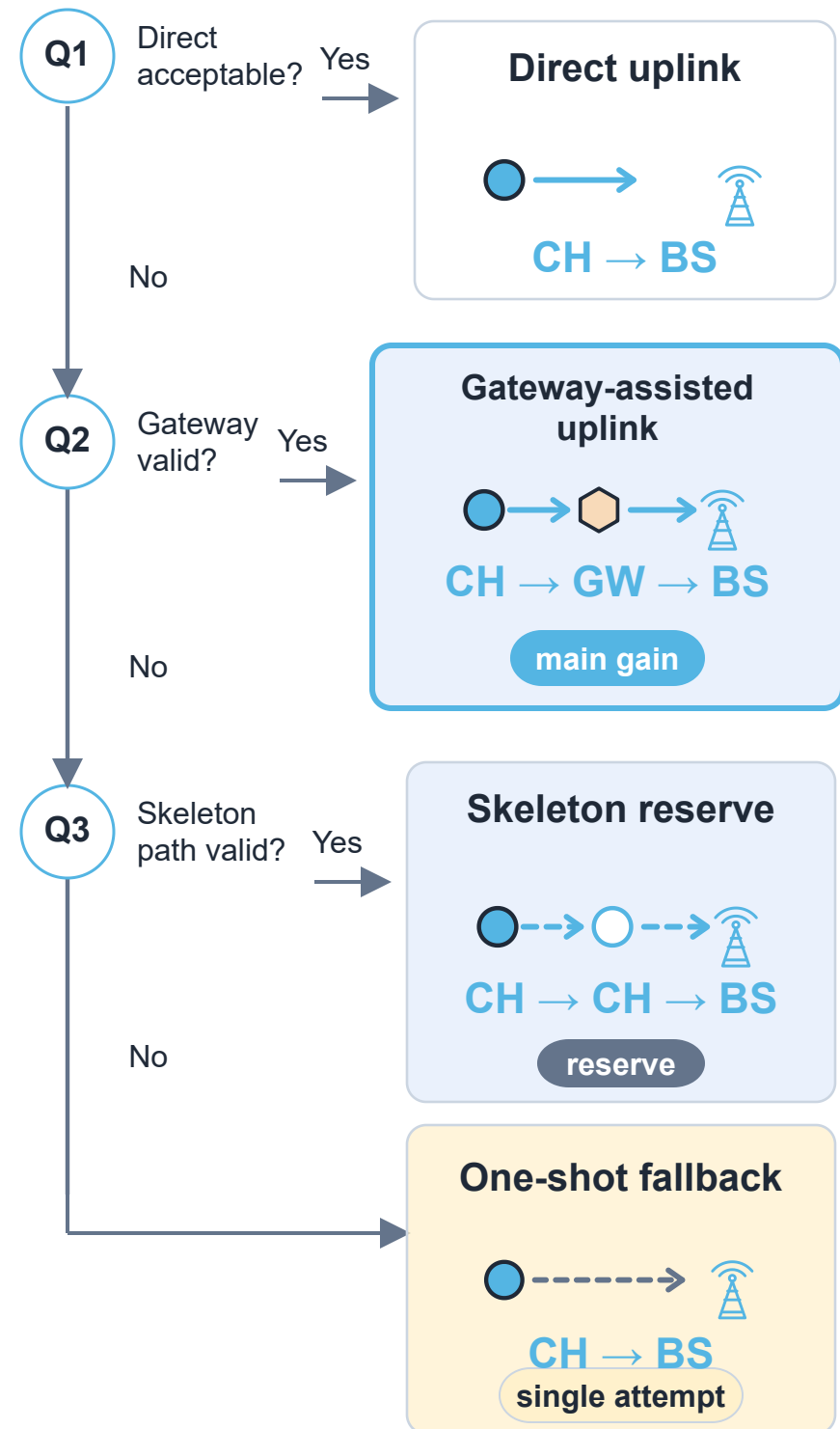


3 CH aggregation



Phase 3 Uplink to BS

pre-transmission route selection



Strict mode: no retransmission; no power step-up